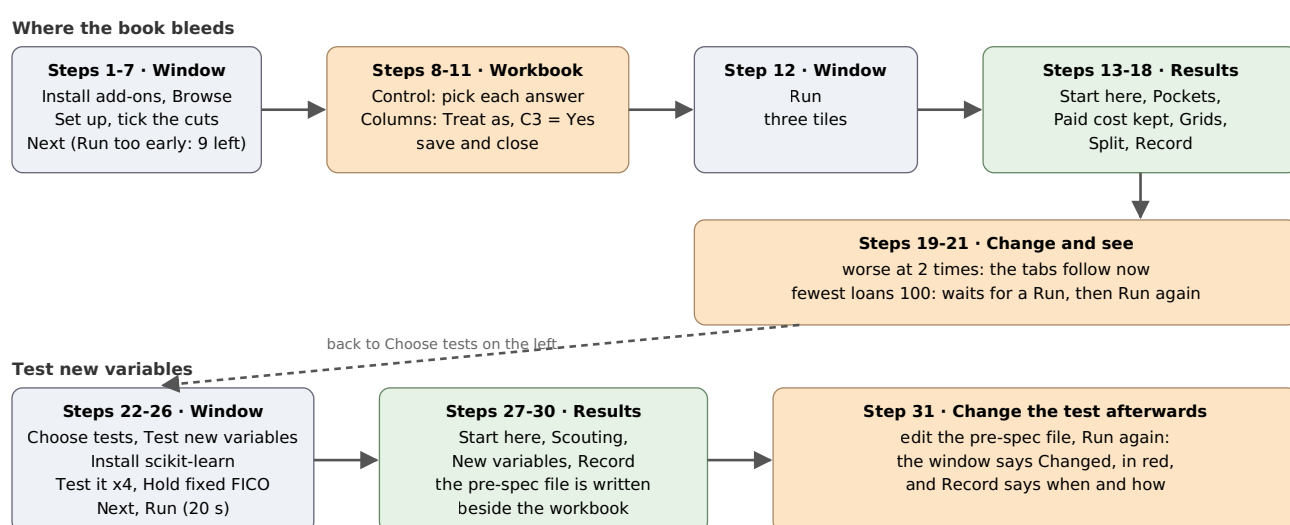


# PocketBook, step by step: finding where a loan book bleeds, and testing a new variable

**Who this is for:** an analyst using PocketBook for the first time. You need the loan file from the bank and Excel. You don't type any commands.

**What you'll have at the end:** a workbook beside the loan file that shows which groups of loans lose more than their share, how much, and whether it could be chance. Then a second kind of run: whether a new column (here, revolving debt) really tells good loans from bad, checked on loans it was never found on.

**Walked on:** 27 Sep 2026, on a made-up book of 8,000 loans with known answers planted in it. Every picture below is the real screen from that walk.



Grey: the PocketBook window. Peach: where you answer (Excel, or the pre-spec file). Green: where you read the results.  
Nothing is typed at a command line. Step 32 is what a Run that stops looks like.

The route: the window, the workbook, the window again, then the results, for each kind of run.

## Words you'll meet

Each is shown the first time on screen. This is what they mean.

| On screen       | What it means                                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Extract</b>  | The loan file from the bank: one row per loan, as a .csv or .xlsx. PocketBook never changes it.                          |
| <b>Workbook</b> | The Excel file PocketBook writes beside the extract. You answer questions in it, and the results land in it.             |
| <b>Band</b>     | A range of a number column, like a FICO score from 496 to 653. <b>Cut into bands</b> means split the column into ranges. |
| <b>Segment</b>  | A group from a category column, like the Broker channel.                                                                 |
| <b>Pocket</b>   | One band crossed with one segment: FICO 496 to 653 <b>and</b> Broker. The thing PocketBook judges.                       |
| <b>Grid</b>     | Every band of one column against every segment of another: one table of pockets.                                         |

| On screen                            | What it means                                                                                                                                              |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GCO</b>                           | Gross charge-offs: the dollars written off on loans that went bad. On screen as <b>Charge-offs</b> .                                                       |
| <b>RANR</b>                          | What the bank kept after losses: interest and fees, less the cost of funds, less charge-offs. On screen as <b>Kept after losses</b> .                      |
| <b>Worse at 1.34x</b>                | A pocket counts as worse when it goes bad at least 1.34 times as often as the loans it is compared with.                                                   |
| <b>p-value</b>                       | How likely a gap this big is by chance alone. Under 5% means it is unlikely to be chance.                                                                  |
| <b>Material</b>                      | Big enough in dollars to matter: at or above the line you pick on Control.                                                                                 |
| <b>Pre-spec</b>                      | A short file that fixes, before the test, exactly what will be tested. It stops anyone moving the goalposts after seeing the answer.                       |
| <b>Development loans / held back</b> | The loans are split by date. The first 70% are where an idea is found ( <b>development</b> ). The last 30% are <b>held back</b> and only used to check it. |
| <b>Scouting</b>                      | The step that looks through the columns you ticked, on the development loans, and proposes which are worth testing.                                        |

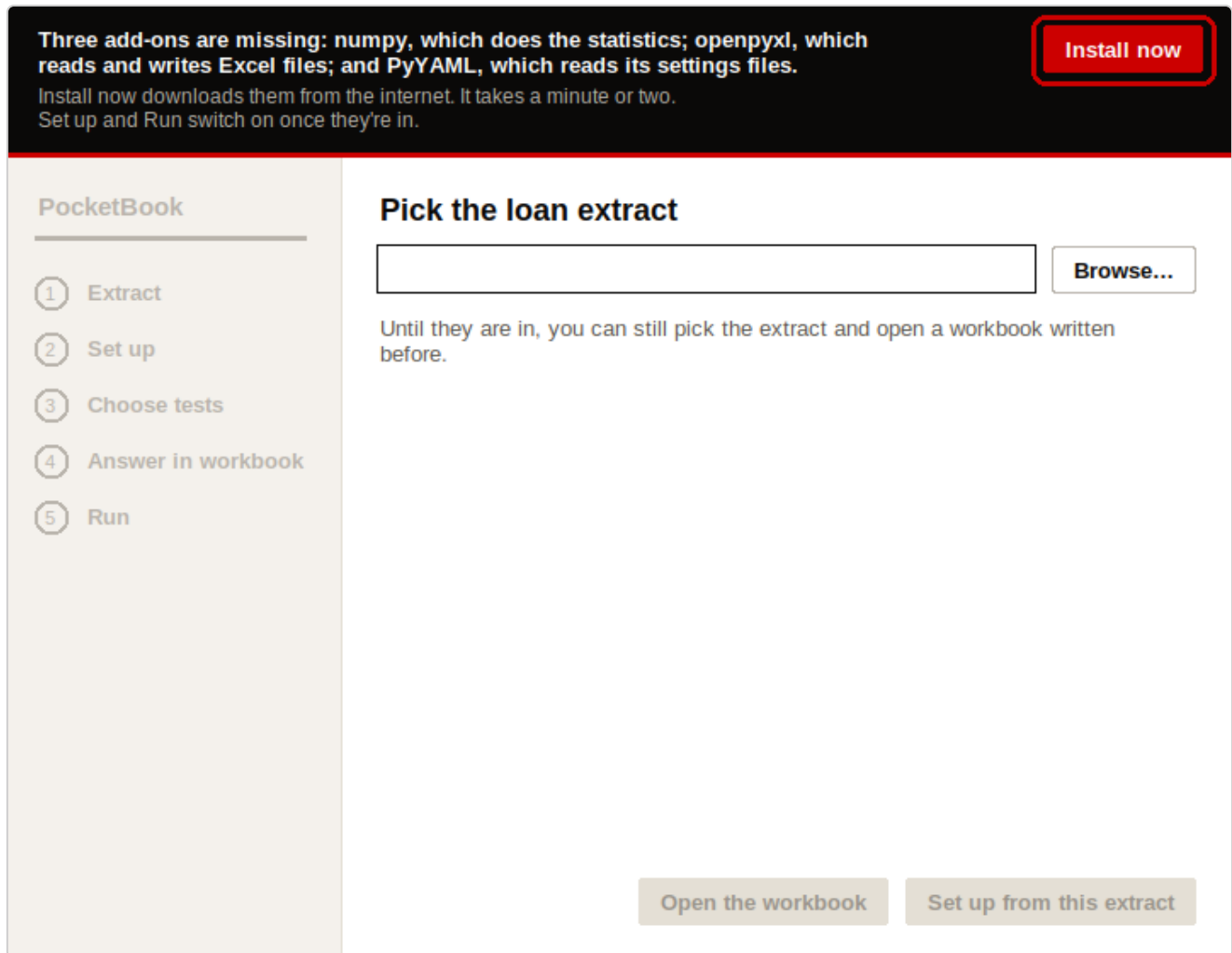
## Part A · Open PocketBook

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### Step 1 · Double-click PocketBook.pyw

The window opens with five steps down the left. The first time, a black bar may say some add-ons are missing. **Add-ons** are free pieces of software PocketBook needs to do its sums and read Excel.

Press **Install now**.



**Correct screen:** the bar names what is missing and what each one does. **Set up** is grey until they're in.

## Step 2 · Pick the extract

Press **Browse...**, find the loan file, and press **Open**.

Three add-ons are missing: numpy, which does the statistics; openpyxl, which reads and writes Excel files; and PyYAML, which reads its settings files.

Install now

Install now downloads them from the internet. It takes a minute or two.  
Set up and Run switch on once they're in.

PocketBook

Pick the loan extract

1 Directory: /home/user/SATC/.claude/worktrees/agent-ae4911b224660d84f

2

3

4

5

|                              |                          |                |
|------------------------------|--------------------------|----------------|
| .claude                      | cashew-institute         | credit-suite   |
| .claude-plugin               | cfpb-mortgage-monitor    | desk           |
| .github                      | client-documents         | docs           |
| bls-laue-county-monitor      | consumer-credit-research | edgar-crit-cl  |
| bureau-credit-risk-dashboard | cowork-plugin            | fdic-peer-mo   |
| canon                        | credit-review-os         | fred-credit-ri |

File name: /tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67-

Files of type: Extracts (\*.csv,\*.xlsx)

Open Cancel

Open the workbook

Set up from this extract

### Step 3 · Wait for the install

It takes a minute or two (a few seconds if the add-ons were fetched before).

PocketBook

1 Extract

Pick the loan file

2 Set up

3 Choose tests

4 Answer in workbook

5 Run

Pick the loan extract

One row per loan, as a .csv or .xlsx. Set up reads it and guesses what each column means. The workbook is written beside it once you've chosen the tests. The extract itself is never changed.

-4db589c3cc31/scratchpad/walk27/book2/Consumer book Q3.csv

Browse...

▼ How Set up recognises columns

A number column with this many values or fewer is a category

12 values ▼

A text column with more values than this is too fine to cut by

50 values ▼

Installed numpy, openpyxl and PyYAML. Everything PocketBook needs is here.

Set up from this extract

**Correct screen:** the black bar has gone. A line says what was installed. **Set up from this extract** is red, so you can press it. The two settings under How Set up recognises columns can be left as they are.

## Part B · Where the book bleeds

### Step 4 · Press Set up from this extract

PocketBook reads the file and guesses what each column is. Nothing is written yet.

page 5 of 30

### PocketBook

- ✓ **Extract**  
Consumer book Q3.csv · 8,000 loans
- ✓ **Set up**  
10 columns read
- 3 Choose tests**  
Where the book bleeds
- ④ Answer in workbook
- ⑤ Run

### What are you running?

Where the book bleeds

Test new variables

| Column      | What it is          | Cut into bands  | Segment by | Split by |
|-------------|---------------------|-----------------|------------|----------|
| LOAN_NBR    | Loan number (key)   |                 |            |          |
| FICO        | FICO score          | ✓               |            | ○        |
| CHANNEL     | Category · 3 values |                 | ✓          |          |
| ORIG_BAL    | Booked amount       | ✓               |            | ○        |
| BAD_FLAG    | Outcome (yes/no)    | · every measure |            |          |
| GCO_AMT     | GCO dollars         | · every measure |            |          |
| RANR_AMT    | RANR dollars        | · every measure |            |          |
| ASSET_CLASS | Category · 4 values |                 | ✓          |          |
| REV_DEBT    | Amount or number    | ✓               |            | ○        |
| ORIG_DATE   | Origination date    |                 |            |          |

This will run: 3 band columns × 2 segment columns = 6 grids, five measures each.

Saved to Control, read-only.

Next: answer in the workbook →

**Correct screen:** the left side shows the file name and "8,000 loans", then "10 columns read". The table lists every column with what PocketBook thinks it is. **Where the book bleeds** is picked at the top right.

The boxes decide what is cut:

- **Cut into bands:** number columns to split into ranges (FICO, ORIG\_BAL, REV\_DEBT are ticked).
- **Segment by:** category columns (CHANNEL, ASSET\_CLASS).
- **Split by:** at most one number column, used to cut every pocket in half at its own middle value.
- The outcome and dollar columns say **every measure**: they are what is measured, not what is cut.

The made-up book has three number columns you can cut. Your bank's file may have more.

## Step 5 · Tick Split by on REV\_DEBT

Press the round button in the **Split by** column on the REV\_DEBT row.

### PocketBook

- ✓ **Extract**  
Consumer book Q3.csv · 8,000 loans
- ✓ **Set up**  
10 columns read
- 3 Choose tests**  
Where the book bleeds
- 4 Answer in workbook
- 5 Run

### What are you running?

Where the book bleeds Test new variables

| Column      | What it is          | Cut into bands | Segment by | Split by        |
|-------------|---------------------|----------------|------------|-----------------|
| LOAN_NBR    | Loan number (key)   |                |            |                 |
| FICO        | FICO score          | ✓              |            |                 |
| CHANNEL     | Category · 3 values |                | ✓          |                 |
| ORIG_BAL    | Booked amount       | ✓              |            |                 |
| BAD_FLAG    | Outcome (yes/no)    |                |            | · every measure |
| GCO_AMT     | GCO dollars         |                |            | · every measure |
| RANR_AMT    | RANR dollars        |                |            | · every measure |
| ASSET_CLASS | Category · 4 values |                | ✓          |                 |
| REV_DEBT    | Amount or number    |                |            |                 |
| ORIG_DATE   | Origination date    |                |            |                 |

This will run: 2 band columns × 2 segment columns = 4 grids, five measures each; split by REV\_DEBT adds 4 more.

Saved to Control, read-only.
Next: answer in the workbook →

**Correct screen:** REV\_DEBT's **Cut into bands** box clears by itself (a column can't be both). The shaded box under the table says what will run: 2 band columns × 2 segment columns = 4 grids ... split by REV\_DEBT adds 4 more.

### Step 6 · Press Next: answer in the workbook

The workbook is written beside the extract: Consumer book Q3 - PocketBook.xlsx.

PocketBook

✓ Extract

Consumer book Q3.csv - 8,000 loans

✓ Set up

10 columns read

✓ Choose tests

Where the book bleeds

4 Answer in workbook

Fill in the shaded cells

5 Run

Answer in the workbook

Consumer book Q3 - PocketBook.xlsx is written beside the extract. Fill in the shaded cells, save and close it, then press Run.

Worked out from this extract: fewest loans 65; worse at 1.34x; better at 0.75x. Each sits beside its setting on Control, for you to take or not.

7 columns to look at first, on the Columns tab.  
Next: fill in the shaded cells on Control and Columns, save, close, and press Run.

Open the workbook

Run

**Correct screen:** step 4 on the left is red. The shaded box gives three values PocketBook worked out from this book (fewest loans in a pocket, how much worse, how much better). They are only suggestions, and each sits beside its question on the Control tab.

### Step 7 · (What happens if you press Run too early)

If you press **Run** now, nothing runs. The window lists every answer still needed, with the tab and cell.

## PocketBook

- ✓ **Extract**  
Consumer book Q3.csv - 8,000 loans
- ✓ **Set up**  
10 columns read
- ✓ **Choose tests**  
Where the book bleeds
- Answer in workbook**  
9 left
- 5 **Run**

### 9 answers needed before Run

Each one opens the workbook at its cell. Nothing ran; your answers so far are kept.

|                    |                                                                               |             |
|--------------------|-------------------------------------------------------------------------------|-------------|
| <b>CONTROL C15</b> | How much worse than its comparison a pocket must be                           | Open at C15 |
| <b>CONTROL C16</b> | How much better than its comparison a pocket must be                          | Open at C16 |
| <b>CONTROL C17</b> | How far profit must move before it counts as more or less                     | Open at C17 |
| <b>CONTROL C18</b> | How sure a difference must be before it counts                                | Open at C18 |
| <b>CONTROL C19</b> | Smallest excess loss worth reporting                                          | Open at C19 |
| <b>CONTROL C20</b> | What a pocket is judged against                                               | Open at C20 |
| <b>CONTROL C24</b> | Fewest loans in a pocket for the usual test (smaller ones get the exact test) | Open at C24 |

Open the workbook
Run

**Correct screen:** "9 answers needed before Run", top to bottom as the workbook shows them. Each has an **Open at** button that opens the workbook on that cell. Step 4 on the left says "9 left". Your answers so far are kept.

## Step 8 · Open the workbook and go to Control

Press **Open the workbook**. The **Control** tab holds the calls only you can make. Every pink cell needs an answer.

### Control

The professional calls. The top block changes results now; the second waits for a Run.

How this tab works  
Changes now  
**Needs a Run**  
Your answer  
Worked out  
Chosen in the launcher  
Materiality levels

The result tabs read these answers with formulas. Change one and every reading, dollar figure, colour and verdict follows at once. The order of rows stays as the last Run left it. These decide which pockets exist and which test each one gets, so they take effect when you press Run again. Status says when an answer differs from what the last Run used. Pick from the list, or type a number under Or your own: a number there wins. A shaded answer still needs one. Comes to shows what the answer amounts to now. Fewest loans, worse at and better at each have a value worked out from this extract, shown beside the setting. It is never picked for you. What you're running and how the pockets are cut. Change them in the launcher's Choose tests, then press Next. They are shown here so a reviewer sees them. For each level: its dollar line, how many pockets have charge-offs above their share that reach it, and their part of all such dollars. It follows your answer live.

| Changes now                                                                   | Needs a Run                                                           | Still needs an answer |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------|
| Changes now - the result tabs follow as you change these                      |                                                                       |                       |
| Setting                                                                       | Your answer to                                                        | Or your own           |
| How much worse than its comparison a pocket must be                           |                                                                       |                       |
| How much better than its comparison a pocket must be                          |                                                                       |                       |
| How far profit must move before it counts as more or less                     |                                                                       |                       |
| How sure a difference must be before it counts                                |                                                                       |                       |
| Smallest excess loss worth reporting                                          |                                                                       |                       |
| What a pocket is judged against                                               |                                                                       |                       |
| To Needs a Run - takes effect when you press Run again                        |                                                                       |                       |
| Setting                                                                       | Your answer to                                                        | Or your own           |
| Fewest loans in a pocket for the usual test (smaller ones get the exact test) |                                                                       |                       |
| Fewest loans with a loss before a loss rate is tested                         |                                                                       |                       |
| How many bands a number column is cut into                                    | 5 bands (recommended)                                                 |                       |
| Where the band edges fall                                                     | Each band holds about the same number of loans (recommended)          |                       |
| How often a real gap should be caught                                         | 80% of the time (recommended)                                         |                       |
| Allowing for testing many pockets at once                                     | Hold down the share of false finds (Benjamini-Hochberg) (recommended) |                       |
| Chosen in the launcher - read-only here                                       |                                                                       |                       |
| What are you running?                                                         | Where the book bleeds                                                 |                       |
| Cut into bands                                                                | FICO, ORIG_BAL                                                        |                       |
| Segment by                                                                    | CHANNEL, ASSET_CLASS                                                  |                       |
| Split every pocket by                                                         | REV_DEBT                                                              |                       |
| A number column with this many values or fewer is a category                  | 12 values                                                             |                       |
| A text column with more values than this is too fine to cut by                | 50 values                                                             |                       |

To change these, go back to Choose tests in the launcher and press Next.

At the right, under **Worked out from the loans**, are the values PocketBook suggests for this book:

[illegible]

The top block (**Changes now**) takes effect in the result tabs as soon as you change it. The second block (**Needs a Run**) takes effect the next time you press Run.

### Step 9 · Answer Control

Pick from each cell's dropdown. On the walk, reading the suggestions:

| Question                                              | Answer picked                                                | Why                                            |
|-------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------|
| How much worse than its comparison a pocket must be   | The smallest significant gap in a typical pocket (suggested) | comes to 1.34 times, worked out from this book |
| How much better ...                                   | The smallest significant gap in a typical pocket (suggested) | 0.75 times                                     |
| How far profit must move before it counts             | Each pocket's own test (suggested)                           |                                                |
| How sure a difference must be before it counts        | 95% sure                                                     |                                                |
| Smallest excess loss worth reporting                  | 1% of the book's total losses                                | came to \$107,354                              |
| What a pocket is judged against                       | The rest of its band                                         |                                                |
| Fewest loans in a pocket for the usual test           | Enough for 5 expected losses (suggested)                     | 65 loans                                       |
| Fewest loans with a loss before a loss rate is tested | 10 losses                                                    |                                                |

The rest already hold a recommended answer.

**Control** The professional calls. The top block changes results now. 40 seconds wait for a Run.

How this tab works  
**Changes now** The result tabs read these answers with formulas. Change one and every reading, dollar figure, colour and verdict follows at once. The order of rows stays as the last Run left it. These decide which pockets exist and which test each one gets, so they take effect when you press Run again. Status says when an answer differs from what the last Run left it. Pick from the list, or type a number under Or your own: a number there wins. A shaded answer still needs one. Comes to show what the answer amounts to now.  
**Your answer** Fewest loans, more at and better at each have a value worked out from this extract, shown beside the setting. It's never picked for you.  
**Worked out** What you're running and how the pockets are cut. Change them in the launcher's Choose tests, then press Next. They are shown here so a reviewer sees them.  
**Chosen in the launcher** For each level: its dollar line, how many pockets have charge-offs above their share that reach it, and their part of all such dollars. It follows your answer live.  
**Materiality levels**

| Changes now                                                                                                                                                                                                                                                                                                                                                                         | Needs a Run                                                                                                                                                                                                                                                               | Still needs an answer                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Changes now - the result tabs follow as you change these</b><br>Setting<br>How much worse than its comparison a pocket must be<br>How much better than its comparison a pocket must be<br>How far profit must move before it counts as more or less<br>How sure a difference must be before it counts<br>Smallest excess loss worth reporting<br>What a pocket is judged against | <b>For all</b><br>The smallest significant gap in a typical pocket (suggested)<br>The smallest significant gap in a typical pocket (suggested)<br>Each pocket's own test (suggested)<br>95% sure<br>1% of the book's total losses<br>The rest of its band                 | <b>For your own</b><br>The smallest significant gap in a typical pocket (suggested)<br>suggested: 1.34x, from this extract<br>suggested: 0.75x, from this extract<br>1% of the book's total losses<br>10% |
| <b>Needs a Run - takes effect when you press Run again</b><br>Setting<br>Fewest loans in a pocket for the usual test (smaller ones get the exact test)<br>Fewest loans with a loss before a loss rate is tested<br>How many bands a number column is cut into<br>Where the band edges fall<br>How often a real gap should be caught<br>Allowing for testing many pockets at once    | <b>For all</b><br>Enough for 5 expected losses (suggested)<br>10 losses<br>5 bands (recommended)<br>Each band holds about the same number of loans (recommended)<br>80% of the time (recommended)<br>Hold down the share of false finds (Benjamin-Hochberg) (recommended) | <b>For your own</b><br>suggested: 65, from this extract                                                                                                                                                   |
| <b>Chosen in the launcher - read only here</b><br>What are you running?<br>Cut into bands<br>Segment by                                                                                                                                                                                                                                                                             | <b>Where the book bleeds</b><br>FICO, ORIG_BAL<br>CHANNEL, ASSET_CLASS                                                                                                                                                                                                    | To change these, go back to Choose tests in the launcher and press Next.                                                                                                                                  |

What each materiality level means, what looks odd in it, and what PocketBook remembers about it.

| Level | What each materiality level means, what looks odd in it, and what PocketBook remembers about it. |
|-------|--------------------------------------------------------------------------------------------------|
| 0.5%  |                                                                                                  |
| 1%    |                                                                                                  |
| 2%    |                                                                                                  |
| 5%    |                                                                                                  |
| 10%   |                                                                                                  |

Charge-offs, more A profit shortfall

**Correct screen:** no pink cell left on Control.

## Step 10 • Answer Columns

On the **Columns** tab:

1. Read each row's **What it is** and **Why we think so**. Fix any that's wrong.
2. Answer each **Odd value** under **Treat as**. On the walk: FICO -9999 on 160 loans is a code for "no score", so **Missing**. RANR negative on 595 loans is real (a loan that charged off loses money), so **Real**.
3. Set **Checked every column?** (C3) to **Yes**.

**Columns** What each column means, what looks odd in it, and what PocketBook remembers about it.

**Checked every column?** ☐ Run won't start until this is Yes.

How this tab works  
**What it is** Set up guesses each column's meaning from its name and values; the reason is beside it. Fix any that's wrong. Confirmed meanings are remembered for the next extract.  
**Odd values** Values that may be codes rather than real numbers: one value far more often than any other, or negatives in a column that's mostly positive. Missing is treated as blank and counted.  
**Treat as** Answer Real or Missing. One left blank is used as recorded.  
**Band edges** Blank means Control's setting. Type edges as 620; 680; 740, or every 20. Look shows what each would cut, live.  
**Remembered** Set Forget? to Yes to drop what was learned about a column at the next Run. It then waits here for you to confirm it again. Answers marked to take effect at the next Run.

| Column      | Samples                           | What it is to     | Why we think so                                           | Blank | Odd values            | Treat as to | Band edges to | Remembered |
|-------------|-----------------------------------|-------------------|-----------------------------------------------------------|-------|-----------------------|-------------|---------------|------------|
| LOAN_NBR    | L0000000, L0000001, L0000002      | Loan number (key) | name contains 'loan'; a different value on every row      | 0%    |                       |             |               | N          |
| FICO        | -9999, 713, 716                   | FICO score        | name contains 'fico'; 98% of values between 300 and 850   | 0%    | -9999 on 160 loans    |             |               | N          |
| CHANNEL     | Online, Broker, Branch            | Category          | 3 different values                                        | 0%    |                       |             |               | N          |
| ORIG_BAL    | 7655.75, 27999.47, 39684.43       | Booked amount     | name contains 'bal'; positive on every loan               | 0%    |                       |             |               | N          |
| BAD_FLAG    | 0, 2, 1                           | Outcome (yes/no)  | name contains 'bad'; 0 and 1 (7.7% are 1)                 | 0%    |                       |             |               | N          |
| GCO_AMT     | 0.0, #N/A, 27620.62               | GCO dollars       | name contains 'gco'; zero on 92% of loans, never negative | 0%    |                       |             |               | N          |
| RANR_AMT    | 3017.35, 3928.24, 5816.98         | RANR dollars      | name contains 'ranr'; numbers                             | 0%    | Negative on 595 loans |             |               | N          |
| ASSET_CLASS | 1, 2, 4                           | Category          | 4 different values                                        | 0%    |                       |             |               | N          |
| REV_DEBT    | 15868.86, 12884.7, 8920.97        | Amount or number  | numbers with 7,889 values                                 | 0%    |                       |             |               | N          |
| ORIG_DATE   | 2021-11-18, 2024-06-24, 2023-07-1 | Origination date  | name contains 'orig'; dates                               | 0%    |                       |             |               | N          |

**Add a column: one divided by another**

Shows here and on Look, and in Choose tests, once you press Set up again. Blank where the bottom is zero or blank.

| New column name to | Top (divided by) | Bottom (divided by) | What it makes |
|--------------------|------------------|---------------------|---------------|
|                    |                  |                     |               |
|                    |                  |                     |               |
|                    |                  |                     |               |

## Step 11 · Save and close the workbook

While it is open in Excel, the window shows a pink bar and **Run** stays grey, because Run can't read answers Excel is still holding.

**PocketBook**

✓ **Extract**  
Consumer book Q3.csv ·  
8,000 loans

✓ **Set up**  
10 columns read

✓ **Choose tests**  
Where the book bleeds

! **Answer in workbook**  
9 left

5 **Run**

### 9 answers needed before Run

Each one opens the workbook at its cell. Nothing ran; your answers so far are kept.

**The workbook is open in Excel.**  
Save and close it first, so Run reads your latest answers.

|                    |                                                           |             |
|--------------------|-----------------------------------------------------------|-------------|
| <b>CONTROL C15</b> | How much worse than its comparison a pocket must be       | Open at C15 |
| <b>CONTROL C18</b> | How sure a difference must be before it counts            | Open at C18 |
| <b>CONTROL C19</b> | Smallest excess loss worth reporting                      | Open at C19 |
| <b>CONTROL C16</b> | How much better than its comparison a pocket must be      | Open at C16 |
| <b>CONTROL C17</b> | How far profit must move before it counts as more or less | Open at C17 |
| <b>CONTROL C20</b> | What a pocket is judged against                           | Open at C20 |

Open the workbook

Run

**Correct screen after closing:** the pink bar goes by itself within a couple of seconds, and **Run** turns red.

## Step 12 · Press Run

On 8,000 loans it took 9 seconds.

PocketBook

✓ Extract

Consumer book Q3.csv - 8,000 loans

✓ Set up

10 columns read

✓ Choose tests

Where the book bleeds

✓ Answer in workbook

All answered

✓ Run

08:23 · 9 seconds

Run finished. Results are in the workbook.

Pockets worse and material

4

charge-offs, of 81

Charge-offs above their share

\$3.09M

in those 4 pockets

Tie-out checks

702 / 702

every grid adds up to the book

Run again

Open at Start here

**Correct screen:** "Run finished." Three tiles: how many pockets are worse on charge-offs and material, the charge-off dollars above their share in them, and the tie-out checks. A **tie-out** adds every grid's pockets back up and checks they come to the book's totals. If one doesn't, the Run stops and says so (see Step 32).

### Step 13 · Read Start here

Press **Open at Start here**.

page 13 of 30

## How this tab works

## Where things stand

What is left before Run, counted live from Control and Columns as you fill them in.

## What the last Run found

Pockets that read worse and are material, and the five largest. Worse? and Material? follow Control; the order and the rest are as of the last Run.

## The tabs

Red tabs you fill in; black tabs hold results; grey tabs are the record.

## WHERE THINGS STAND

Answers still needed

0

Columns to confirm

0

Odd values to answer

0

Changes waiting for a Run

0

Control

Columns

Columns - Treat as

Control - Needs a Run

## WHAT THE LAST RUN FOUND

Pockets worse and material, charge-offs

4 of 81

Dollars above their share, in those

\$3,094,991

Pockets keeping less, worse and material

4 short \$3,248,654

Tie-out checks

702 of 702 agree

| Largest, worse and material | Segment       | Loans | × its comparison | Dollars above share | Worse? | Material? |
|-----------------------------|---------------|-------|------------------|---------------------|--------|-----------|
| FICO 496 - 653              | Broker        | 523   | 2.62×            | \$1,494,129         | Yes    | Yes       |
| FICO 496 - 653              | ASSET_CLASS 4 | 392   | 1.74×            | \$643,768           | Yes    | Yes       |
| ORIG_BAL 37,951 - 49,151    | ASSET_CLASS 4 | 414   | 1.75×            | \$529,198           | Yes    | Yes       |
| FICO 712 - 745              | Online        | 564   | 2.06×            | \$427,896           | Yes    | Yes       |

Every pocket, every measure: the Pockets tab.

## THE TABS

You answer

[Control: the professional calls](#)[Columns: meanings, odd values, memory](#)[Look: each number column's shape](#)

Results

[Pockets: every pocket, worse first](#)[Paid, cost, kept: paid against cost](#)[Grids: one grid at a time, and how common](#)[Split: each pocket halved](#)

Record

[Record: what ran, the tie-outs, every Run](#)

**Correct screen: Where things stand** is all zeros (nothing left to answer). **What the last Run found** gives the same numbers as the window, then the largest pockets that are worse and material. On the walk the top one was FICO 496 to 653 / Broker, 2.62 times its band's charge-offs, \$1,494,129 above its share. That is the pocket planted in the made-up book.

## Step 14 · Read Pockets

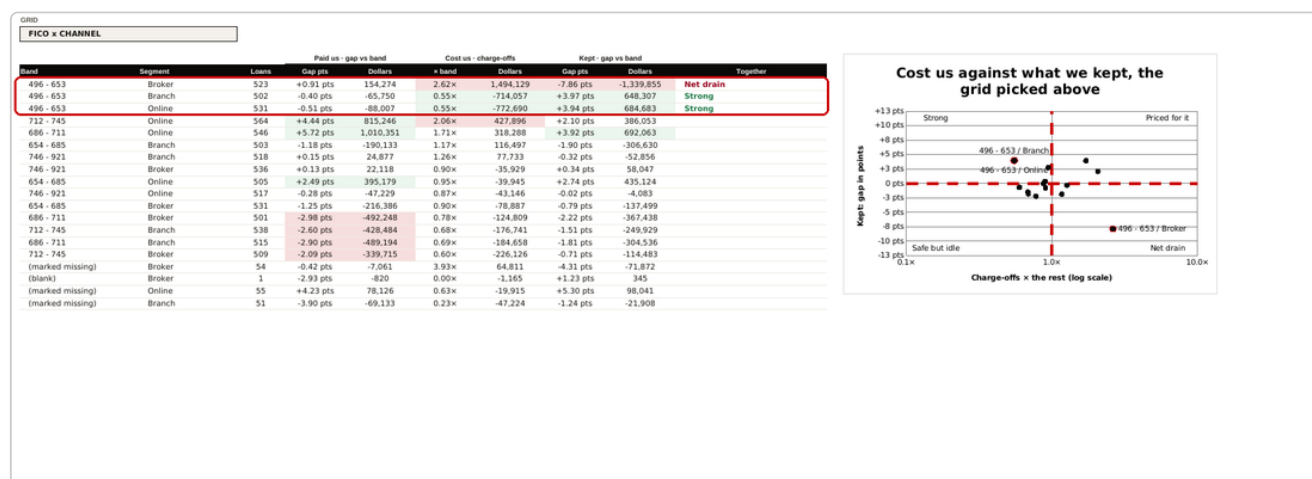
Every pocket, worst first. Pick the **Measure** (Bad loans, Bad dollars, Charge-offs, Kept after losses, Earned before losses), **Pockets** (two-way, or split) and **Show** at the top.

| Pockets                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |               |       |             |              |                |                       |          |             |           |                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------|-------|-------------|--------------|----------------|-----------------------|----------|-------------|-----------|-------------------|
| Which pockets lose more than their share, how much, and whether it's real and big enough to matter                                                                                                                                                                                                                                                                                                                                  |                          |               |       |             |              |                |                       |          |             |           |                   |
| How this tab works                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |               |       |             |              |                |                       |          |             |           |                   |
| A pocket                                                                                                                                                                                                                                                                                                                                                                                                                            |                          |               |       |             |              |                |                       |          |             |           |                   |
| One band of one column crossed with one segment of another, in every grid chosen in the launcher. Loans: how many are in it. Split by REV_DEBT: each pocket halved at its own median REV_DEBT, each half tested like any other pocket.                                                                                                                                                                                              |                          |               |       |             |              |                |                       |          |             |           |                   |
| Rest of band                                                                                                                                                                                                                                                                                                                                                                                                                        |                          |               |       |             |              |                |                       |          |             |           |                   |
| Every other loan in the same band, judged against on Control picks it or the rest of the book (every other loan in the book); now the rest of its band. It decides the gap, the excess and both verdicts. A pocket alone in its band has nothing beside it to compare with, so it is judged against the rest of the book, whatever Control says.                                                                                    |                          |               |       |             |              |                |                       |          |             |           |                   |
| Excess                                                                                                                                                                                                                                                                                                                                                                                                                              |                          |               |       |             |              |                |                       |          |             |           |                   |
| What the pocket lost above what it would have lost at the rest's rate: bad loans for Bad loans, dollars for the dollar measures. For Kept after losses and Earned before losses, the dollars it fell short of the rest, and the gap is in points of booked dollars.                                                                                                                                                                 |                          |               |       |             |              |                |                       |          |             |           |                   |
| Worse?                                                                                                                                                                                                                                                                                                                                                                                                                              |                          |               |       |             |              |                |                       |          |             |           |                   |
| Yes: at least 1.34x the rest (for profit, short of it by the profit line on Control), with a p-value under 5%. Not sure: the gap is there, but the p-value isn't under it. Too few losses: fewer than 10 (a setting for the next Run), so not tested. No: anything else.                                                                                                                                                            |                          |               |       |             |              |                |                       |          |             |           |                   |
| p-value                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |               |       |             |              |                |                       |          |             |           |                   |
| The chance of a gap at least this big if the pocket were no different from the rest, after the allowance for testing many pockets at once (Benjamini-Hochberg, across one grid and one measure). Bad loans: the z test, or the exact test for a pocket under 65 loans. Dollar measures: the loans are shuffled 10,000 times, within the band for the rest of its band, and it is how often a gap as big turned up. Under 5% counts. |                          |               |       |             |              |                |                       |          |             |           |                   |
| Material?                                                                                                                                                                                                                                                                                                                                                                                                                           |                          |               |       |             |              |                |                       |          |             |           |                   |
| Yes when the excess reaches the materiality line on Control (the Material at tile, in the measure's own unit). It is judged apart from Worse?: a pocket can be one without the other.                                                                                                                                                                                                                                               |                          |               |       |             |              |                |                       |          |             |           |                   |
| Could have caught                                                                                                                                                                                                                                                                                                                                                                                                                   |                          |               |       |             |              |                |                       |          |             |           |                   |
| The smallest gap a pocket this size would catch 80% of the time at the last Run's confidence. As of the last Run.                                                                                                                                                                                                                                                                                                                   |                          |               |       |             |              |                |                       |          |             |           |                   |
| Order                                                                                                                                                                                                                                                                                                                                                                                                                               |                          |               |       |             |              |                |                       |          |             |           |                   |
| Sorted as of the last Run: worse pockets first, largest excess first, then the rest by their excess. Last come pockets losing more only against the other comparison, so that changing judged against on Control finds them here. A change on Control updates Worse?, Material?, the dollars and the colours at once, not the order. The caption counts what the dropdowns pick.                                                    |                          |               |       |             |              |                |                       |          |             |           |                   |
| Holds FICO fixed?                                                                                                                                                                                                                                                                                                                                                                                                                   |                          |               |       |             |              |                |                       |          |             |           |                   |
| Split only. REV_DEBT moves with FICO (correlation -0.53), so a split pocket in a grid that doesn't hold FICO fixed may be showing FICO, not REV_DEBT. Those rows come last, in grey, with no verdict colour.                                                                                                                                                                                                                        |                          |               |       |             |              |                |                       |          |             |           |                   |
| LINES IN USE NOW                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |               |       |             |              |                |                       |          |             |           |                   |
| live from Control                                                                                                                                                                                                                                                                                                                                                                                                                   |                          |               |       |             |              |                |                       |          |             |           |                   |
| Worse at                                                                                                                                                                                                                                                                                                                                                                                                                            |                          |               |       |             |              |                |                       |          |             |           |                   |
| 1.34×                                                                                                                                                                                                                                                                                                                                                                                                                               |                          |               |       |             |              |                |                       |          |             |           |                   |
| Better at                                                                                                                                                                                                                                                                                                                                                                                                                           |                          |               |       |             |              |                |                       |          |             |           |                   |
| 0.75×                                                                                                                                                                                                                                                                                                                                                                                                                               |                          |               |       |             |              |                |                       |          |             |           |                   |
| How sure                                                                                                                                                                                                                                                                                                                                                                                                                            |                          |               |       |             |              |                |                       |          |             |           |                   |
| 95%                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |               |       |             |              |                |                       |          |             |           |                   |
| Material at                                                                                                                                                                                                                                                                                                                                                                                                                         |                          |               |       |             |              |                |                       |          |             |           |                   |
| 6.2 bad loans                                                                                                                                                                                                                                                                                                                                                                                                                       |                          |               |       |             |              |                |                       |          |             |           |                   |
| Judged against                                                                                                                                                                                                                                                                                                                                                                                                                      |                          |               |       |             |              |                |                       |          |             |           |                   |
| Rest of its band                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |               |       |             |              |                |                       |          |             |           |                   |
| Order and "Could have caught" are from the last Run, 2026-09-27 08:23.                                                                                                                                                                                                                                                                                                                                                              |                          |               |       |             |              |                |                       |          |             |           |                   |
| MEASURE                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |               |       |             |              |                |                       |          |             |           |                   |
| POCKETS                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |               |       |             |              |                |                       |          |             |           |                   |
| SHOW                                                                                                                                                                                                                                                                                                                                                                                                                                |                          |               |       |             |              |                |                       |          |             |           |                   |
| 7 worse and material · 7 worse · 41 shown                                                                                                                                                                                                                                                                                                                                                                                           |                          |               |       |             |              |                |                       |          |             |           |                   |
| Bad loans                                                                                                                                                                                                                                                                                                                                                                                                                           |                          |               |       |             |              |                |                       |          |             |           |                   |
| Two-way                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |               |       |             |              |                |                       |          |             |           |                   |
| All                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |               |       |             |              |                |                       |          |             |           |                   |
| #                                                                                                                                                                                                                                                                                                                                                                                                                                   | Band                     | Segment       | Loans | This pocket | Rest of band | × rest of band | Bad loans above share | Worse?   | p-value     | Material? | Could have caught |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                   | FICO 496 - 653           | Broker        | 523   | 24.47%      | 10.27%       | 2.38×          | 74.3                  | Yes      | under 0.01% | Yes       | 1.46×             |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                   | FICO 712 - 745           | Online        | 564   | 8.69%       | 3.63%        | 2.39×          | 28.5                  | Yes      | 0.02%       | Yes       | 1.45×             |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                   | FICO 496 - 653           | ASSET_CLASS 4 | 392   | 20.41%      | 13.24%       | 1.54×          | 28.1                  | Yes      | 1.4%        | Yes       | 1.54×             |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                   | ORIG_BAL 37,951 - 49,151 | ASSET_CLASS 4 | 414   | 12.32%      | 7.00%        | 1.76×          | 22.0                  | Yes      | 1.5%        | Yes       | 1.52×             |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                   | FICO 654 - 685           | ASSET_CLASS 4 | 410   | 11.71%      | 7.26%        | 1.61×          | 18.2                  | Yes      | 4.5%        | Yes       | 1.52×             |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                   | ORIG_BAL 26,324 - 37,950 | ASSET_CLASS 4 | 412   | 9.95%       | 5.82%        | 1.71×          | 17.0                  | Yes      | 4.3%        | Yes       | 1.52×             |
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                   | FICO 712 - 745           | ASSET_CLASS 4 | 427   | 8.20%       | 4.39%        | 1.87×          | 16.2                  | Yes      | 3.4%        | Yes       | 1.51×             |
| 8                                                                                                                                                                                                                                                                                                                                                                                                                                   | ORIG_BAL 15,449 - 26,323 | Broker        | 537   | 11.73%      | 8.09%        | 1.45×          | 19.6                  | Not sure | 11.0%       | Yes       | 1.46×             |
| 9                                                                                                                                                                                                                                                                                                                                                                                                                                   | ORIG_BAL 49,152 - 59,984 | Broker        | 520   | 8.46%       | 5.46%        | 1.55×          | 15.6                  | Not sure | 11.0%       | Yes       | 1.47×             |
| 10                                                                                                                                                                                                                                                                                                                                                                                                                                  | FICO 686 - 711           | Online        | 546   | 7.69%       | 5.31%        | 1.45×          | 13.0                  | Not sure | 16.0%       | Yes       | 1.46×             |
| 11                                                                                                                                                                                                                                                                                                                                                                                                                                  | ORIG_BAL 15,449 - 26,323 | ASSET_CLASS 4 | 413   | 11.38%      | 8.59%        | 1.32×          | 11.5                  | No       | 23.3%       | Yes       | 1.52×             |
| 12                                                                                                                                                                                                                                                                                                                                                                                                                                  | ORIG_BAL 5,005 - 15,448  | ASSET_CLASS 4 | 425   | 9.65%       | 6.98%        | 1.38×          | 11.3                  | Not sure | 21.9%       | Yes       | 1.51×             |
| 13                                                                                                                                                                                                                                                                                                                                                                                                                                  | ORIG_BAL 49,152 - 59,984 | ASSET_CLASS 4 | 384   | 8.59%       | 5.76%        | 1.49×          | 10.9                  | Not sure | 19.3%       | Yes       | 1.54×             |
| 14                                                                                                                                                                                                                                                                                                                                                                                                                                  | FICO 686 - 711           | ASSET_CLASS 4 | 407   | 8.11%       | 5.45%        | 1.49×          | 10.8                  | Not sure | 16.6%       | Yes       | 1.53×             |
| 15                                                                                                                                                                                                                                                                                                                                                                                                                                  | ORIG_BAL 37,951 - 49,151 | Broker        | 531   | 9.23%       | 7.95%        | 1.16×          | 6.8                   | No       | 86.7%       | Yes       | 1.46×             |
| 16                                                                                                                                                                                                                                                                                                                                                                                                                                  | ORIG_BAL 26,324 - 37,950 | ASSET_CLASS 4 | 412   | 9.95%       | 5.82%        | 1.71×          | 17.0                  | Yes      | 4.3%        | Yes       | 1.52×             |
| 17                                                                                                                                                                                                                                                                                                                                                                                                                                  | FICO 313 - 348           | ASSET_CLASS 4 | 428   | 8.30%       | 4.50%        | 1.83×          | 14.3                  | Yes      | 3.4%        | Yes       | 1.51×             |
| #                                                                                                                                                                                                                                                                                                                                                                                                                                   | Band                     | Segment       | Loans | This pocket | Rest of band | × rest of band | Bad loans above share | Worse?   | p-value     | Material? | Could have caught |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                   | FICO 496 - 653           | Broker        | 523   | 24.47%      | 10.27%       | 2.38×          | 74.3                  | Yes      | under 0.01% | Yes       | 1.46×             |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                   | FICO 712 - 745           | Online        | 564   | 8.69%       | 3.63%        | 2.39×          | 28.5                  | Yes      | 0.02%       | Yes       | 1.45×             |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                   | FICO 496 - 653           | ASSET_CLASS 4 | 392   | 20.41%      | 13.24%       | 1.54×          | 28.1                  | Yes      | 1.4%        | Yes       | 1.54×             |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                   | ORIG_BAL 37,951 - 49,151 | ASSET_CLASS 4 | 414   | 12.32%      | 7.00%        | 1.76×          | 22.0                  | Yes      | 1.5%        | Yes       | 1.52×             |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                   | FICO 654 - 685           | ASSET_CLASS 4 | 410   | 11.71%      | 7.26%        | 1.61×          | 18.2                  | Yes      | 4.5%        | Yes       | 1.52×             |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                   | ORIG_BAL 26,324 - 37,950 | ASSET_CLASS 4 | 412   | 9.95%       | 5.82%        | 1.71×          | 17.0                  | Yes      | 4.3%        | Yes       | 1.52×             |
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                   | FICO 712 - 745           | ASSET_CLASS 4 | 427   | 8.20%       | 4.39%        | 1.87×          | 16.2                  | Yes      | 3.4%        | Yes       | 1.51×             |
| 8                                                                                                                                                                                                                                                                                                                                                                                                                                   | ORIG_BAL 15,449 - 26,323 | Broker        | 537   | 11.73%      | 8.09%        | 1.45×          | 19.6                  | Not sure | 11.0%       | Yes       | 1.46×             |
| 9                                                                                                                                                                                                                                                                                                                                                                                                                                   | ORIG_BAL 49,152 - 59,984 | Broker        | 520   | 8.46%       | 5.46%        | 1.55×          | 15.6                  | Not sure | 11.0%       | Yes       | 1.47×             |
| 10                                                                                                                                                                                                                                                                                                                                                                                                                                  | FICO 686 - 711           | Online        | 546   | 7.69%       | 5.31%        | 1.45×          | 13.0                  | Not sure | 16.0%       | Yes       | 1.46×             |
| 11                                                                                                                                                                                                                                                                                                                                                                                                                                  | ORIG_BAL 15,449 - 26,323 | ASSET_CLASS 4 | 413   | 11.38%      | 8.59%        | 1.32×          | 11.5                  | No       | 23.3%       | Yes       | 1.52×             |
| 12                                                                                                                                                                                                                                                                                                                                                                                                                                  | ORIG_BAL 5,005 - 15,448  | ASSET_CLASS 4 | 425   | 9.65%       | 6.98%        | 1.38×          | 11.3                  | Not sure | 21.9%       | Yes       | 1.51×             |
| 13                                                                                                                                                                                                                                                                                                                                                                                                                                  | ORIG_BAL 49,152 - 59,984 | ASSET_CLASS 4 | 384   | 8.59%       | 5.76%        | 1.49×          | 10.9                  | Not sure | 19.3%       | Yes       | 1.54×             |
| 14                                                                                                                                                                                                                                                                                                                                                                                                                                  | FICO 686 - 711           | ASSET_CLASS 4 | 407   | 8.11%       | 5.45%        | 1.49×          | 10.8                  | Not sure | 16.6%       | Yes       | 1.53×             |
| 15                                                                                                                                                                                                                                                                                                                                                                                                                                  | ORIG_BAL 37,951 - 49,151 | Broker        | 531   | 9.23%       | 7.95%        | 1.16×          | 6.8                   | No       | 86.7%       | Yes       | 1.46×             |
| 16                                                                                                                                                                                                                                                                                                                                                                                                                                  | ORIG_BAL 26,324 - 37,950 | ASSET_CLASS 4 | 412   | 9.95%       | 5.82%        | 1.71×          | 17.0                  | Yes      | 4.3%        | Yes       | 1.52×             |
| 17                                                                                                                                                                                                                                                                                                                                                                                                                                  | FICO 313 - 348           | ASSET_CLASS 4 | 428   | 8.30%       | 4.50%        | 1.83×          | 14.3                  | Yes      | 3.4%        | Yes       | 1.51×             |

**Correct screen:** the **Lines in use now** bar repeats your Control answers. **Worse?** is Yes, Not sure (a gap, but it could be chance) or No. **Material?** is Yes when the excess reaches your dollar line.

## Step 15 • Read Paid, cost, kept

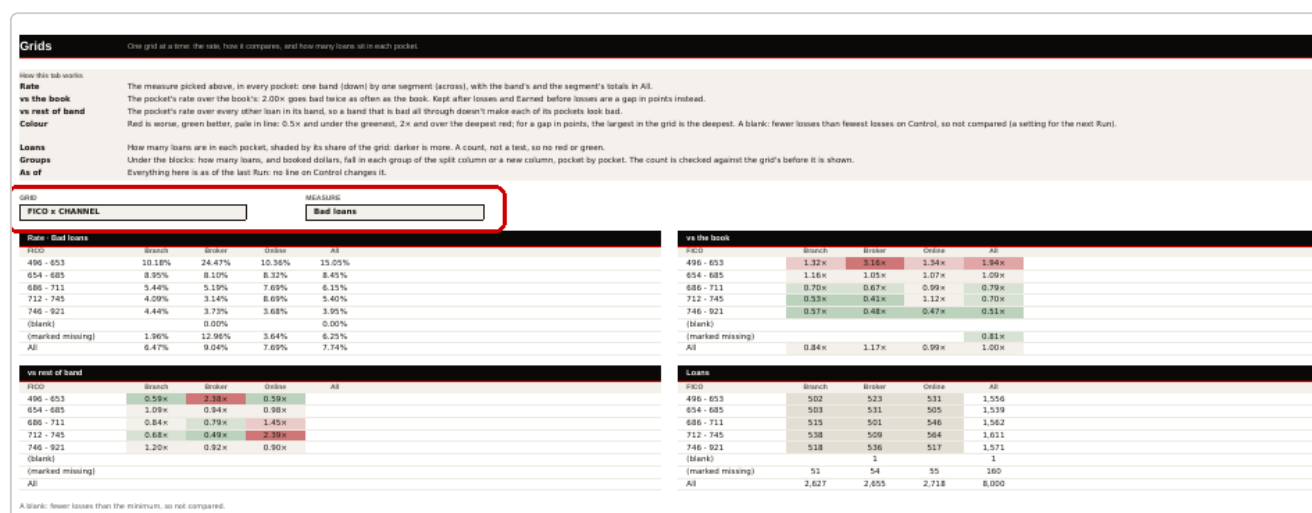
What each pocket paid the bank, what its losses cost, and what was kept. **Together** reads both sides at once: Net drain is more charge-offs and less kept; Strong is fewer charge-offs and more kept.



**Correct screen:** FICO 496 to 653 / Broker reads **Net drain**. The chart puts each pocket by its charge-offs (across) and what it kept (up).

## Step 16 • Read Grids

One grid at a time: the rate in each pocket, the rate against the whole book, against the rest of its band, and how many loans each pocket holds. Pick the grid and measure at the top.



## Step 17 • Read Split

Every pocket cut in half at its own middle REV\_DEBT. Does the high half do worse?

Split

Every pocket split in two at its own median REV\_DEBT. does the high half do worse?

How this tab works

What it does

Inside each pocket the loans are sorted by REV\_DEBT and cut at that pocket's own median. The high half is compared with the low half, so the band and segment are the same on both sides. REV\_DEBT isn't cut into bands of its own while it splits.

High vs low

The high half's rate over the low half's: 2.00x means the high half goes bad, or loses, twice as often. Kept after losses and Earned before losses are a gap in points instead: +0.30 pts means the high half keeps 0.30 points more per booked dollar.

High vs low, all

The high halves' actual total against what it would be at their low halves' rates, added over every pocket, with its range at 95% sure. For Bad loans only, the odds are pooled too (Mantel-Haenszel: a standard way to combine pockets without mixing their loans), with their p-value (Cochran-Mantel-Haenszel), and Cochran's Q asks whether the gap is the same size in every pocket.

p-value

The chance of a gap at least this big if the two halves were no different. Under 5% counts, and is bold. Bad loans: the gap in standard errors; a dollar measure: the loans dealt into the two halves at random inside their pocket, 10,000 times. The pockets are after the allowance for many tests (Benjamini-Hochberg, across one grid and one measure); a gap that is not significant is in brackets, unshaded. The summary is one pooled test per grid and measure, with no allowance. A blank: a half has fewer loans or losses than the minimums (65 loans, 10 losses).

What it holds

A grid holds fixed only its band and segment. REV\_DEBT moves with FICO (correlation -0.53), so in a grid that doesn't hold FICO fixed part of every gap may be FICO, not REV\_DEBT: the chip says which. Grids that hold it fixed come first in the list. Loans are treated as independent of each other.

As of

The numbers are the last Run's. The range, the brackets and the bold follow the confidence on Control.

LINES IN USE NOW

live from Control

Worse at

Better at

How sure

Material at

Judged against

The numbers are from the last Run, 2026-09-27 08:23.

1.34x

0.75x

95%

\$107,354

Rest of its band

GRID

FICO x CHANNEL

Holds FICO fixed

This grid holds FICO fixed (REV\_DEBT's correlation with it: -0.53). It doesn't hold ORIG\_BAL fixed (REV\_DEBT's correlation with it: -0.02).

| Measure              | Pockets | High half worse in | High vs low, all | Range (95% sure)   | p-value     | As odds | p-value, as odds |
|----------------------|---------|--------------------|------------------|--------------------|-------------|---------|------------------|
| Bad loans            | 15      | 14 of 15           | 2.02x            | 1.79x to 2.25x     | under 0.01% | 2.19x   | under 0.01%      |
| Bad dollars          | 15      | 15 of 15           | 2.14x            | 1.87x to 2.40x     | under 0.01% |         |                  |
| Charge-offs          | 15      | 14 of 15           | 2.13x            | 1.86x to 2.41x     | under 0.01% |         |                  |
| Kept after losses    | 15      | 13 of 15           | -2.78 pts        | -3.65 to -1.92 pts | under 0.01% |         |                  |
| Earned before losses | 15      | 5 of 15            | +0.26 pts        | -0.21 to +0.72 pts | 29.2%       |         |                  |

Same in every pocket? Bad loans: no sign the gap differs between pockets; Bad dollars, Charge-offs, Kept after losses, Earned before losses: not tested: dollar rate.

MEASURE

Bad loans

Bad loans, high vs low

|                  | Branch  | Broker  | Online  |
|------------------|---------|---------|---------|
| 496 - 653        | 2.39x   | 2.67x   | (1.63x) |
| 654 - 685        | (1.37x) | 2.92x   | 2.24x   |
| 686 - 711        | (2.12x) | (1.90x) | 3.20x   |
| 712 - 745        | (2.14x) | (0.78x) | (1.58x) |
| 746 - 921        | (1.30x) | (2.33x) | (1.12x) |
| (blank)          |         |         |         |
| (marked missing) |         |         |         |

p-value per pocket

|                  | Branch | Broker      | Online |
|------------------|--------|-------------|--------|
| 496 - 653        | 0.76%  | under 0.01% | 12.8%  |
| 654 - 685        | 33.5%  | 0.40%       | 2.9%   |
| 686 - 711        | 12.7%  | 14.3%       | 0.31%  |
| 712 - 745        | 13.6%  | 66.1%       | 14.3%  |
| 746 - 921        | 60.3%  | 12.8%       | 80.9%  |
| (blank)          |        |             |        |
| (marked missing) |        |             |        |

**Correct screen:** on Bad loans, the high half is worse in 14 of 15 pockets, 2.02 times overall. The grid holds FICO fixed, which matters because REV\_DEBT moves with FICO. The tab says so beside the grid.

## Step 18 · Read Record

What ran, whether it adds up, what was left out, and every Run. This is the tab for a reviewer.

| Record                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What ran, whether it adds up, what was left out, and every Run. For the reviewer.                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p>How this tab works</p> <p><b>What it holds</b></p> <p><b>Settings</b></p> <p><b>Every Run</b></p>                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p>What the last Run was, whether it adds up, which tests it used, what it left out, and every Run and refusal. Nothing here is a result: the result tabs hold these.</p> <p>Each Control setting: the answer in use now, and what the last Run used. A setting changed since the Run is shaded: the result tabs follow a Changes-now setting at once, and a Needs-a-Run one at the next Run.</p> <p>Newest first. Each entry is what that Run used, and a refusal says what it was waiting for.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| This Run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Settings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Item                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Default                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Extract                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Consumer book Q3.csv                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Loans run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 8,000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| What was run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Where the book bleeds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Record of this run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Consumer book Q3 - PocketBook - what ran.yaml, beside this workbook: every setting the run used, kept for the file. It is replaced by the next Run.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Origination dates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2021-06-30 to 2026-03-30 (8,000 loans; 0 without a readable date)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Band edges used: FICO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 654; 686; 712; 746 (5 bands)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Band edges used: ORIG_BAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 15,449; 26,324; 37,951; 49,152 (5 bands)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Split                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | REV_DEBT, each pocket halved at its own median. REV_DEBT isn't cut on its own while it splits. Split pockets:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 158.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| How closely REV_DEBT moves with FICO                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | correlation -0.53                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| How closely REV_DEBT moves with ORIG_BAL                                                                                                                                                                                                                                                                                                                                                                                                                                                             | correlation -0.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Does it add up                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Tests used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Test                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Tie-out checks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Outcome, share of loans: the z test, pooled, for a pocket of 65 loans or more, and the exact test (Fisher's) below that. Every dollar rate: the loans are shuffled 10,000 times, within the band for the rest of its band, and its p-value is how often a shuffle made a gap as big. Profit and contribution are compared as a gap in points, never a multiple. The split's odds: Cochran-Mantel-Haenszel, which asks whether an odds ratio this far from 1 could come from shuffling loans within their pockets. It has no continuity correction: nothing is taken off the gap between actual and expected before it is squared. |
| Worse now: Outcome, share of loans                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | p-value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Worse now: Outcome, share of booked dollars                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Standard error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Worse now: GCO per booked dollar                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Worse now: Profit after losses: RANR per booked dollar                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Worse now: Contribution before losses per booked dollar                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Loans needed for a 1.34x gap: Outcome, share of loans                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Loans needed for a 1.34x gap: Outcome, share of booked dollars                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Loans needed for a 1.34x gap: GCO per                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Step 19 · Change a "Changes now" answer and watch the tabs follow

On Control, change **How much worse than its comparison a pocket must be to 2 times**. Save.

PocketBook

Consumer book Q3.csv · 8,000 loans · 10 columns · last Run 2026-09-27 08:23

How this tab works

Where things stand

What the last Run found

The tabs

What is left before Run, counted live from Control and Columns as you fill them in.

Pockets that read worse and are material, and the five largest. Worse? and Material? follow Control; the order and the rest are as of the last Run.

WHERE THINGS STAND

Answers still needed

Columns to confirm

Odd values to answer

Changes waiting for a Run

0

0

0

0

Control

Columns

Columns · Treat as

Control · Needs a Run

WHAT THE LAST RUN FOUND

Pockets worse and material, charge-offs

Dollars above their share, in those

Pockets keeping less, worse and material

Tie-out checks

2 of 81

\$1,922,025

4 short \$3,248,654

702 of 702 agree

Largest, worse and material

Segment

Loans

× its comparison

Dollars above share

Worse?

Material?

FICO 496 - 653

Broker

523

2.62×

\$1,494,129

Yes

Yes

FICO 496 - 653

ASSET\_CLASS 4

392

1.74×

\$643,768

No

Yes

ORIG\_BAL 37,951 - 49,151

ASSET\_CLASS 4

414

1.75×

\$529,198

No

Yes

FICO 712 - 745

Online

564

2.06×

\$427,896

Yes

Yes

Every pocket, every measure: the Pockets tab

**Correct screen:** without pressing Run, Start here now says **2 of 81** (it said 4), and Pockets' **Worse?** column follows. Rows now under 2 times read **No**:

| Live from Control |                          | 2.00x         | 0.75x | 95%                                       | 6.2 bad loans | Rest of its band |                       |        |             |           |                   |
|-------------------|--------------------------|---------------|-------|-------------------------------------------|---------------|------------------|-----------------------|--------|-------------|-----------|-------------------|
| MEASURE           |                          | POCKETS       |       | SHOW                                      |               |                  |                       |        |             |           |                   |
| Bad loans         |                          | Two-way       | All   | 2 worse and material - 2 worse - 41 shown |               |                  |                       |        |             |           |                   |
| #                 | Band                     | Segment       | Loans | This pocket                               | Rest of band  | x rest of band   | Bad loans above share | Worse? | p-value     | Material? | Could have caught |
| 1                 | FICO 496 - 653           | Broker        | 523   | 24.47%                                    | 10.27%        | 2.38x            | 74.3                  | Yes    | under 0.01% | Yes       | 1.46x             |
| 2                 | FICO 712 - 745           | Online        | 564   | 8.69%                                     | 3.63%         | 2.39x            | 28.5                  | Yes    | 0.02%       | Yes       | 1.45x             |
| 3                 | FICO 496 - 653           | ASSET_CLASS 4 | 392   | 20.41%                                    | 13.24%        | 1.54x            | 28.1                  | No     | 1.4%        | Yes       | 1.54x             |
| 4                 | ORIG_BAL 37,951 - 49,151 | ASSET_CLASS 4 | 414   | 12.32%                                    | 7.80%         | 1.76x            | 22.0                  | No     | 1.5%        | Yes       | 1.52x             |
| 5                 | FICO 654 - 685           | ASSET_CLASS 4 | 410   | 11.71%                                    | 7.26%         | 1.61x            | 18.2                  | No     | 4.5%        | Yes       | 1.52x             |
| 6                 | ORIG_BAL 26,324 - 37,950 | ASSET_CLASS 4 | 412   | 9.95%                                     | 5.82%         | 1.71x            | 17.0                  | No     | 4.3%        | Yes       | 1.52x             |
| 7                 | FICO 712 - 745           | ASSET_CLASS 4 | 427   | 8.20%                                     | 4.39%         | 1.87x            | 16.2                  | No     | 3.4%        | Yes       | 1.51x             |
| 8                 | ORIG_BAL 15,449 - 26,323 | Broker        | 537   | 11.73%                                    | 8.09%         | 1.45x            | 19.6                  | No     | 11.0%       | Yes       | 1.46x             |
| 9                 | ORIG_BAL 49,152 - 59,984 | Broker        | 520   | 8.46%                                     | 5.46%         | 1.55x            | 15.6                  | No     | 11.0%       | Yes       | 1.47x             |

## Step 20 • Change a "Needs a Run" answer and see it wait

On Control, change **Fewest loans in a pocket for the usual test** to **100 loans**. Save.

**PocketBook**
Consumer book Q3.csv · 8,000 loans · 10 columns · last Run 2026-09-27 08:23

How this tab works

**Where things stand** What is left before Run, counted live from Control and Columns as you fill them in.

**What the last Run found** Pockets that read worse and are material, and the five largest. Worse? and Material? follow Control; the order and the rest are as of the last Run.

**The tabs** Red tabs you fill in; black tabs hold results; grey tabs are the record.

**WHERE THINGS STAND**

Answers still needed Columns to confirm Odd values to answer

0 0 0

Control Columns Columns · Treat as

Changes waiting for a Run 1

Control · Needs a Run

1 change is waiting for a Run. Fewest loans in a pocket for the usual test (smaller ones get the exact test): Enough for 5 expected losses (suggested) → 100 loans (Control C24). The result tabs still show the last Run. Save, close, and press Run in the launcher.

**WHAT THE LAST RUN FOUND**

Pockets worse and material, charge-offs Dollars above their share, in those Pockets keeping less, worse and material Tie-out checks

2 of 81 \$1,922,025 4 short \$3,248,654 702 of 702 agree

| Largest, worse and material | Segment       | Loans | x its comparison | Dollars above share | Worse? | Material? |
|-----------------------------|---------------|-------|------------------|---------------------|--------|-----------|
| FICO 496 - 653              | Broker        | 523   | 2.62x            | \$1,494,129         | Yes    | Yes       |
| FICO 496 - 653              | ASSET_CLASS 4 | 392   | 1.74x            | \$643,768           | No     | Yes       |
| ORIG_BAL 37,951 - 49,151    | ASSET_CLASS 4 | 414   | 1.75x            | \$529,198           | No     | Yes       |
| FICO 712 - 745              | Online        | 564   | 2.06x            | \$427,896           | Yes    | Yes       |

Every pocket, every measure: the Pockets tab.

**Correct screen:** Start here's **Changes waiting for a Run** says 1, in red, and a pink line names the change and its cell. On Control its **Status** reads **Waiting for a Run**:

How far profit must move before it counts as more or less

How sure a difference must be before it counts

Smallest excess loss worth reporting

What a pocket is judged against

Each pocket's own test (suggested)

95% sure

1% of the book's total losses

The rest of its band

each pocket's own test

95% sure

1% of the book's total losses

The rest of its band

Needs a Run - takes effect when you press Run again

Setting

Fewest loans in a pocket for the usual test (smaller ones get the exact test)

Fewest loans with a loss before a loss rate is tested

How many bands a number column is cut into

Where the band edges fall

How often a real gap should be caught

Allowing for testing many pockets at once

Your answer to

100 loans

10 losses (recommended)

5 bands (recommended)

Each band holds about the same number of loans (recommended)

80% of the time (recommended)

Hold down the share of false finds (Benjamin-Hochberg) (recommended)

On your run

65 (worked out from this book)

10 losses

5 bands

Each band holds about the same number of loans

80% of the time

Hold down the share of false finds (Benjamin-Hochberg)

Last Run used

65 (worked out from this book)

10 losses

5 bands

Each band holds about the same number of loans

80% of the time

Hold down the share of false finds (Benjamin-Hochberg)

Changes waiting for a Run suggested: 65, from this extract at the last Run

Same as last Run

Same as last Run

Same as last Run

Same as last Run

Same as last Run

Chosen in the launcher - read only here

What are you running?

Where the book bleeds

Cut into bands

Segment by

FICO, ORIG\_BAL

CHANNEL, ASSET\_CLASS

The result tabs still show the last Run until you press Run. Each says "1 Control change waits for a Run" beside **Lines in use now**.

## Step 21 · Close the workbook and press Run again

PocketBook

✓ Extract

Consumer book Q3.csv · 8,000 loans

✓ Set up

10 columns read

✓ Choose tests

Where the book bleeds

✓ Answer in workbook

All answered

✓ Run

08:26 · 9 seconds

Run finished. Results are in the workbook.

Pockets worse and material

2

charge-offs, of 81

Charge-offs above their share

\$1.92M

in those 2 pockets

Tie-out checks

702 / 702

every grid adds up to the book

Run again

Open at Start here

**Correct screen:** "Run finished." The waiting count on Start here is back to 0.

## Part C · Test new variables

A **new variable** is a column you think might tell good loans from bad, beyond what is already known. Here, revolving debt, with the FICO score held fixed so that REV\_DEBT isn't just standing in for FICO.

## Step 22 · Go back to Choose tests and pick Test new variables

Press **Choose tests** on the left, then **Test new variables** at the top right.

### PocketBook

- ✓ **Extract**  
Consumer book Q3.csv · 8,000 loans
- ✓ **Set up**  
10 columns read
- 3 Choose tests**  
Test new variables
- 4 Answer in workbook
- 5 Run

## What are you running?

Where the book bleeds **Test new variables**

| Column      | What it is          | Outcome | Test it                  | Hold fixed               |
|-------------|---------------------|---------|--------------------------|--------------------------|
| LOAN_NBR    | Loan number (key)   |         |                          |                          |
| FICO        | FICO score          |         | <input type="checkbox"/> | <input type="checkbox"/> |
| CHANNEL     | Category · 3 values |         | <input type="checkbox"/> | <input type="checkbox"/> |
| ORIG_BAL    | Booked amount       |         | <input type="checkbox"/> | <input type="checkbox"/> |
| BAD_FLAG    | Outcome (yes/no)    | •       |                          |                          |
| GCO_AMT     | GCO dollars         |         |                          |                          |
| RANR_AMT    | RANR dollars        |         |                          |                          |
| ASSET_CLASS | Category · 4 values |         | <input type="checkbox"/> | <input type="checkbox"/> |
| REV_DEBT    | Amount or number    |         | <input type="checkbox"/> | <input type="checkbox"/> |
| ORIG_DATE   | Origination date    |         |                          |                          |

Find on 70% confirm on the rest
Or confirm a saved shortlist **Browse...**

**Install scikit-learn**

One add-on is missing: scikit-learn, which finds new variables (scouting). Press Install scikit-learn, or confirm a saved shortlist instead.

Saved to Control, read-only.
Next: answer in the workbook →

**Correct screen:** the columns now read **Outcome**, **Test it** and **Hold fixed**. If a red button says **Install scikit-learn**, press it. **scikit-learn** is the free add-on that does the scouting.

## Step 23 · Wait for scikit-learn

**PocketBook**

- ✓ **Extract**  
Consumer book Q3.csv · 8,000 loans
- ✓ **Set up**  
10 columns read
- 3 Choose tests**  
Test new variables
- 4 Answer in workbook
- 5 Run

### What are you running?

Where the book bleeds **Test new variables**

| Column      | What it is          | Outcome | Test it                  | Hold fixed               |
|-------------|---------------------|---------|--------------------------|--------------------------|
| LOAN_NBR    | Loan number (key)   |         |                          |                          |
| FICO        | FICO score          |         | <input type="checkbox"/> | <input type="checkbox"/> |
| CHANNEL     | Category · 3 values |         | <input type="checkbox"/> | <input type="checkbox"/> |
| ORIG_BAL    | Booked amount       |         | <input type="checkbox"/> | <input type="checkbox"/> |
| BAD_FLAG    | Outcome (yes/no)    |         |                          |                          |
| GCO_AMT     | GCO dollars         |         |                          |                          |
| RANR_AMT    | RANR dollars        |         |                          |                          |
| ASSET_CLASS | Category · 4 values |         | <input type="checkbox"/> | <input type="checkbox"/> |
| REV_DEBT    | Amount or number    |         | <input type="checkbox"/> | <input type="checkbox"/> |
| ORIG_DATE   | Origination date    |         |                          |                          |

Find on **70%** confirm on the rest Or confirm a saved shortlist **Browse...**

Tick at least one input to test.

Installed scikit-learn. Everything PocketBook needs is here.

Saved to Control, read-only. **Next: answer in the workbook →**

**Correct screen:** a line says it was installed. The box under the table asks you to tick at least one input.

## Step 24 · Tick what to test and what to hold fixed

Tick **Test it** on REV\_DEBT, ASSET\_CLASS, CHANNEL and ORIG\_BAL. Tick **Hold fixed** on FICO. Leave **Find on** at **70%**: the first 70% of the loans by date are used to find, the rest to confirm.

### PocketBook

- ✓ **Extract**  
Consumer book Q3.csv · 8,000 loans
- ✓ **Set up**  
10 columns read
- 3 Choose tests**  
Test new variables
- 4 Answer in workbook
- 5 Run

## What are you running?

Where the book bleeds    Test new variables

| Column      | What it is          | Outcome | Test it                             | Hold fixed                          |
|-------------|---------------------|---------|-------------------------------------|-------------------------------------|
| LOAN_NBR    | Loan number (key)   |         |                                     |                                     |
| FICO        | FICO score          |         | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| CHANNEL     | Category · 3 values |         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| ORIG_BAL    | Booked amount       |         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| BAD_FLAG    | Outcome (yes/no)    | 🎯       |                                     |                                     |
| GCO_AMT     | GCO dollars         |         |                                     |                                     |
| RANR_AMT    | RANR dollars        |         |                                     |                                     |
| ASSET_CLASS | Category · 4 values |         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| REV_DEBT    | Amount or number    |         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| ORIG_DATE   | Origination date    |         |                                     |                                     |

Find on 70% confirm on the rest    Or confirm a saved shortlist Browse...

This will run: 4 inputs (REV\_DEBT, ASSET\_CLASS, CHANNEL, ORIG\_BAL) against BAD\_FLAG, each with and without FICO held fixed: 8 tests, found on 70% and confirmed on 30%.

Installed scikit-learn. Everything PocketBook needs is here.

Saved to Control, read-only.

Next: answer in the workbook →

**Correct screen:** 4 inputs ... against BAD\_FLAG, each with and without FICO held fixed: 8 tests, found on 70% and confirmed on 30%.

## Step 25 · Press Next

### PocketBook

- ✓ **Extract**  
Consumer book Q3.csv - 8,000 loans
- ✓ **Set up**  
10 columns read
- ✓ **Choose tests**  
Test new variables
- 4** **Answer in workbook**  
Fill in the shaded cells
- 5 **Run**

### Answer in the workbook

Consumer book Q3 - PocketBook.xlsx is written beside the extract. Fill in the shaded cells, save and close it, then press Run.

2 columns to look at first, on the Columns tab.  
Everything is answered. Press Run.

[Open the workbook](#) [Run](#)

**Correct screen:** "Everything is answered. Press Run." Your Control answers carry over. Control now shows only the questions this kind of run uses, and **What are you running?** reads Finding and testing a new variable:

Control

The professional calls. The top block changes results now, the second waits for a Run.

How this tab works

Changes now

Needs a Run

Your answer

Worked out

Chosen in the launcher

Materiality levels

The result tabs read these answers with formulas. Change one and every reading, dollar figure, colour and verdict follows at once. The order of rows stays the same. These decide which pockets exist and which test each one gets, so they take effect when you press Run again. Status says when an answer differs from Pick from the list, or type a number under Or your own: a number there wins. A shaded answer still needs one. Comes to shows what the answer amounts to. Worse at has a value worked out from this extract, from the groups being confirmed, shown beside the setting. It is never picked for you. What you're running and how the pockets are cut. Change them in the launcher's Choose tests, then press Next. They are shown here so a reviewer sees Material? on New variables compares each group's excess with the level you pick, live. The panel of levels counts pockets, so it is hidden on this run, when you press Run.

|             |             |       |
|-------------|-------------|-------|
| Changes now | Needs a Run | Still |
|-------------|-------------|-------|

Changes now - the result tabs follow as you change these

| Setting                                             | Your answer                   | Or your own | Comes to        |
|-----------------------------------------------------|-------------------------------|-------------|-----------------|
| How much worse than its comparison a pocket must be | 2 Times                       |             | 2.00x           |
| How sure a difference must be before it counts      | 95% sure                      |             |                 |
| Smallest excess loss worth reporting                | 1% of the book's total losses |             | \$107,354 1% of |

Needs a Run - takes effect when you press Run again

| Setting                                    | Your answer to                                                        | Or your own             |
|--------------------------------------------|-----------------------------------------------------------------------|-------------------------|
| How many bands a number column is cut into | 5 bands (recommended)                                                 |                         |
| Where the band edges fall                  | Each band holds about the same number of loans (recommended)          | n/a Each band holds     |
| Allowing for testing many pockets at once  | Hold down the share of false finds (Benjamini-Hochberg) (recommended) | n/a Hold down the share |

Chosen in the launcher - read-only here

To change these, go back to Choose tests in the launcher and press Next

|                                                                |                                          |
|----------------------------------------------------------------|------------------------------------------|
| What are you running?                                          | Finding and testing a new variable       |
| Scout first, or test from a pre-spec already written?          | Scout first                              |
| Cut into bands                                                 | FICO                                     |
| Segment by                                                     | None                                     |
| Split every pocket by                                          |                                          |
| Tested against                                                 | BAD_FLAG                                 |
| Inputs tested                                                  | REV_DEBT, ASSET_CLASS, CHANNEL, ORIG_BAL |
| Held fixed                                                     | FICO                                     |
| Found on                                                       | 70%, confirmed on the rest               |
| Saved shortlist (pre-spec file)                                |                                          |
| A number column with this many values or fewer is a category   | 12 values                                |
| A text column with more values than this is too fine to cut by | 50 values                                |

## Step 26 • Press Run

On 8,000 loans it took 20 seconds. PocketBook scouts on the development loans, writes the pre-spec Consumer book Q3 - pre-spec.yaml beside the workbook, and only then tests it on the held-back loans.

### PocketBook

- ✓ **Extract**  
Consumer book Q3.csv - 8,000 loans
- ✓ **Set up**  
10 columns read
- ✓ **Choose tests**  
Test new variables
- ✓ **Answer in workbook**  
All answered
- ✓ **Run**  
09:06 - 20 seconds

### Run finished. Results are in the workbook.

REV\_DEBT groups worse than 11,000 - 14,999  
**1 of 2**  
on the holdout, 95% sure

Their share of bad loans  
**54%**  
on the holdout

Follows the pre-spec  
**Written now**  
by this Run's scouting

Run again

Open at Start here

**Correct screen:** the first tile names the column and its reference group (the middle group, which every other is compared with): **1 of 2** REV\_DEBT groups worse, on the held-back loans. The last tile says the pre-spec was **written now, by this Run's scouting**. It can't differ from itself, so it doesn't say "Yes".

## Step 27 · Read Start here

### PocketBook

Consumer book Q3.csv - 8,000 loans - 10 columns - last Run 2026-09-27 09:03

How this tab works

**Where things stand** What is left before Run, counted live from Control and Columns as you fill them in.

**What the last Run found** Each group of the tested column on the holdout, against the reference group. Significant? follows the confidence on Control; the rest is as of the last Run.

**The tabs** Red tabs you fill in; black tabs hold results; grey tabs are the record.

**WHERE THINGS STAND**

|                      |                    |                      |                           |
|----------------------|--------------------|----------------------|---------------------------|
| Answers still needed | Columns to confirm | Odd values to answer | Changes waiting for a Run |
| 0                    | 0                  | 0                    | 0                         |
| Control              | Columns            | Columns · Treat as   | Control · Needs a Run     |

**WHAT THE LAST RUN FOUND**

|                                                   |                                        |                               |                                     |
|---------------------------------------------------|----------------------------------------|-------------------------------|-------------------------------------|
| Groups worse than 11,000 - 14,999, on the holdout | Their share of the holdout's bad loans | Loans tested                  | Follows the pre-spec                |
| 1 of 2                                            | 54%                                    | 5,604 found - 2,395 held back | Written now: by this Run's scouting |

| Group of REV_DEBT           | Loans | Bad rate | x 11,000 - 14,999's odds | p-value     | Significant? | Share of bad loans |
|-----------------------------|-------|----------|--------------------------|-------------|--------------|--------------------|
| 0 - 10,999                  | 1,131 | 5.31%    | 1.24x                    | 38.4%       | No           | 32%                |
| 11,000 - 14,999 (reference) | 546   | 4.76%    | 1.00x                    |             |              | 14%                |
| 15,000 - 48,522             | 718   | 14.07%   | 2.69x                    | under 0.01% | Yes, worse   | 54%                |

[Found beside confirmed, with and without the columns held fixed, and how it's worked out: the New variables tab.](#)

**Correct screen:** one row per REV\_DEBT group on the held-back loans. 15,000 and up is **Yes**, worse: 2.69 times the odds of going bad, holding 54% of the bad loans.

## Step 28 • Read Scouting

Which candidates the book leans on, ranked, on the development loans only.

Scouting

Which of the 4 candidates does the book lean on, and where does each one bend? On the development loans only.

How this tab works

What it is

Found on

Candidates

The forest

Importance

With and without

Noise floor

Proposed?

Suggested bins

Reference

Correlated with

The pre-spec

As of

Outcome

Candidates - proposed

Held fixed

Noise floor

Development loans

BAD\_FLAG

4 • 1 proposed

FICO

0.0255

5,604 • 432 bad

Worked out at the last Run, 2026-09-27 08:29, on the development loans. Ranked by the larger of the two importances.

Candidate

Rank

Importance

FICO held fixed

Suggested bins

Reference

Correlated with

Proposed?

Why

REV\_DEBT

1

0.1540

0.0791

0 - 10,999; 11,000 - 14,999; 15,000 - 47,478

11,000 - 14,999

Yes

clears the noise floor

CHANNEL

2

0.0144

0.0075

Each value

No

a category

ASSET\_CLASS

3

0.0080

0.0054

Each value

No

a category

ORIG\_BAL

4

-0.0011

0.0030

5,024 - 6,199; 6,200 - 59,984

6,200 - 59,984

No

below the noise floor

Held fixed, for comparison: their own importance with the candidates

Column

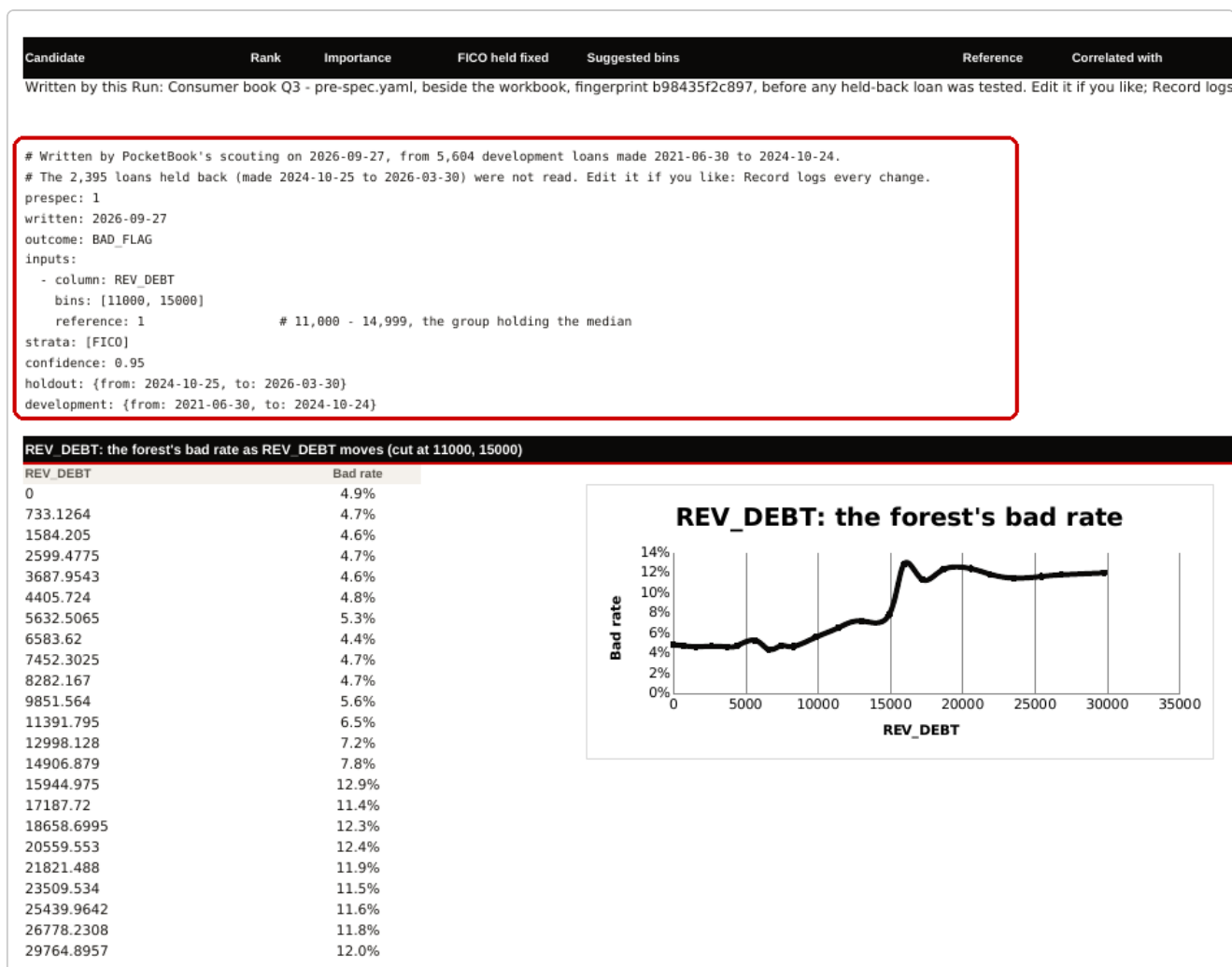
Importance

FICO

0.0526

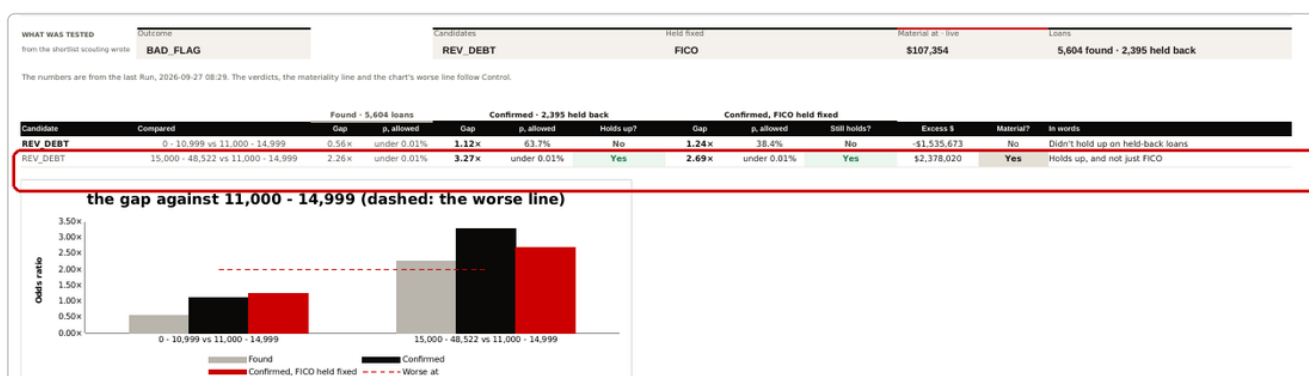
**Correct screen:** REV\_DEBT first, **Proposed? Yes**, above the **noise floor** (what a column scores by pure chance). CHANNEL and ASSET\_CLASS are categories, ranked but not proposed. ORIG\_BAL is below the floor.

Further down, the pre-spec as written, and REV\_DEBT's curve with the suggested cuts:



## Step 29 • Read New variables

Does REV\_DEBT still tell good loans from bad on loans it was never found on, and once FICO is held fixed?



**Correct screen:** 15,000 and up: **Holds up? Yes** and **Still holds? Yes** (with FICO held fixed), **Material? Yes**. Under 11,000: found on development, but it **didn't hold up** on the held-back loans.

# Step 30 · Read Record

Record

What ran, whether it adds up, what was left out, and every Run. For the reviewer.

How this tab works

What it holds

Settings

Every Run

What the last Run was, whether it adds up, which tests it used, what it left out, and every Run and refusal. Nothing here is a result: the result tabs hold those.

Each Control setting: the answer in use now, and what the last Run used. A setting changed since the Run is shaded; the result tabs follow a Changes now setting at once, and a Needs a Run one at the next Run.

Newest first. Each entry is what that Run used, and a refusal says what it was waiting for.

This Run

Item

Detail

Extract

Consumer book Q3.csv

Loans run

8,000

What was run

Finding and testing a new variable: scout first

Record of this run

Consumer book Q3 - PocketBook - what ran.yaml, beside this workbook: every setting the run used, kept for the file. It is replaced by the next Run.

Origination dates

2021-06-30 to 2026-03-30 (8,000 loans; 0 without a readable date)

Band edges used: FICO

654; 686; 712; 746 (5 bands)

Scouting

4 candidates on 5,604 development loans made 2021-06-30 to 2024-10-24; 1 proposed: REV\_DEBT. See the Scouting tab.

Scouting held back

2,395 loans made 2024-10-25 to 2026-03-30: none of their outcomes or values was read by scouting.

Scouting's pre-spec

Written by this Run: Consumer book Q3 - pre-spec.yaml, beside the workbook, fingerprint b98435f2c897, before any held-back loan was tested. Edit it if you like. Record logs every change.

Pre-spec

/tmp/claude-Qi-home-user-SATC/b4305af6-8769-5da1-bd67-4db589c3c313/scratchpad/walk27/book/Consumer book Q3 - pre-spec.yaml

Pre-spec commit

not in a git repository, so not committed: a pre-spec only counts once it is committed

Pre-spec fingerprint

b98435f2c897: the first 12 characters of the file's SHA-256, so any change to it shows

What the pre-spec says

written: 2026-09-27

outcome: BAD\_FLAG

inputs: 1

1. column: REV\_DEBT

1. bins: 11,000, 15,000 (groups: 0 - 10,999; 11,000 - 14,999; 15,000 - 48,522)

1. reference: 11,000 - 14,999

strata: FICO

confidence: 0.95

holdout: 2024-10-25 to 2026-03-30

development: 2021-06-30 to 2024-10-24

Warning

This run doesn't count as the pre-specified one until the pre-spec is committed, unchanged.

Differs from the pre-spec

nowhere: this run used what it says

Holdout

2,395 of this extract's loans were made in the holdout (2024-10-25 to 2026-03-30), the first on 2024-10-25 and the last on 2026-03-30. This run touched the holdout.

Confirmatory test

Development 2021-06-30 to 2024-10-24: 5,604 loans, 432 bad; holdout 2024-10-25 to 2026-03-30: 2,395 loans, 187 bad. See the New variables tab. Left out of this test only (the loan counts on Record include them): 1 with no readable outcome.

Runs that touched the holdout

1 on Record's Every Run, this one included

This pre-spec's held-back runs

In order: this Run (b98435f2c897).

Does it add up

Check

Result

Bleed tabs

None: testing a new variable runs only the confirmatory test. Any left by an earlier Run were taken off.

Settings

Setting

In use now

Last Run used

How many bands a number column is cut into

5 bands

5 bands

Where the band edges fall

Each band holds about the same number of loans

Each band holds about the same number of loans

Smallest excess less worth reporting

1.0% of the book's GCO, \$107,354

1.0% of the book's GCO, \$107,354

How much worse than its comparison a pocket must be

worse at 2.00x

worse at 2.00x

How sure a difference must be before it counts

95% sure

95% sure

Allowing for testing many pockets at once

Hold down the share of false finds (Benjamini-Hochberg)

Hold down the share of false finds (Benjamini-Hochberg)

Tests used

Where

Test

p-value

The chance of a gap at least this big if there were no real difference, after the allowance for many tests. Below 5% is significant, at 95% sure. Two-sided: a gap either way counts.

**Correct screen:** the pre-spec's fingerprint (a short code that changes if the file changes by even one character), and one run that touched the held-back loans.

# Step 31 · Edit the pre-spec after the held-back run, and Run again

Open Consumer book Q3 - pre-spec.yaml in Notepad. Change bins: [11000, 15000] to bins: [10000, 16000]. Save. Press **Run again**.

This is what the pre-spec exists to catch: changing the test after seeing the answer. PocketBook runs it, and labels it.

PocketBook

✓ Extract

Consumer book Q3.csv - 8,000 loans

✓ Set up

10 columns read

✓ Choose tests

Test new variables

✓ Answer in workbook

All answered

✓ Run

09:07 · 20 seconds

Run finished. Results are in the workbook.

REV\_DEBT groups worse than 10,000 - 15,999

1 of 2

on the holdout, 95% sure

Their share of bad loans

45%

on the holdout

Follows the pre-spec

Changed

after a held-back run: see Record

Run again

Open at Start here

**Correct screen:** the last tile says **Changed, after a held-back run**, in red. On Record, under **This Run**, a red **Warning** gives the date of the earlier run and both fingerprints:

|                                |                                                                                                                                                                                                                                                 |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | confidence: 0.99                                                                                                                                                                                                                                |
|                                | holdout: 2024-10-25 to 2026-03-30                                                                                                                                                                                                               |
|                                | development: 2021-06-30 to 2024-10-24                                                                                                                                                                                                           |
| Warning                        | This run doesn't count as the pre-specified one until the pre-spec is committed, unchanged.                                                                                                                                                     |
| Warning                        | Changed pre-spec: Consumer book Q3 - pre-spec.yaml changed after the held-back run of 2026-09-27 08:29: fingerprint b98435f2c897 then, 5fe791ff8df7 now. This run is on the changed pre-spec.                                                   |
| Differs from the pre-spec      | nowhere: this run used what it says                                                                                                                                                                                                             |
| Holdout                        | 2,395 of this extract's loans were made in the holdout (2024-10-25 to 2026-03-30), the first on 2024-10-25 and the last on 2026-03-30. This run touched the holdout.                                                                            |
| Confirmatory test              | Development 2021-06-30 to 2024-10-24: 5,604 loans, 432 bad; holdout 2024-10-25 to 2026-03-30: 2,395 loans, 187 bad. See the New variables tab. Left out of this test only (the loan counts on Record include them): 1 with no readable outcome. |
| Runs that touched the holdout  | 2 on Record's Every Run, this one included                                                                                                                                                                                                      |
| This pre-spec's held-back runs | In order: 2026-09-27 08:29 (b98435f2c897); this run (5fe791ff8df7, changed).                                                                                                                                                                    |

## When PocketBook says no

| What you did                                | What the window says                                | What to do                                                                               |
|---------------------------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------|
| Opened it before the add-ons were installed | Black bar: "Three add-ons are missing ..." (Step 1) | Press <b>Install now</b> . If it fails, press <b>Copy for IT</b> and send them the note. |
| Pressed Run before answering (Step 7)       | "9 answers needed before Run", each with its cell   | Press <b>Open at</b> on each, answer, save, close, Run.                                  |

| What you did                                             | What the window says                                                        | What to do                                                                |
|----------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Left the workbook open in Excel (Step 11)                | Pink bar: "The workbook is open in Excel." Run is grey.                     | Save and close it. The bar goes by itself.                                |
| Picked Test new variables without scikit-learn (Step 22) | "One add-on is missing: scikit-learn ..."                                   | Press <b>Install scikit-learn</b> , or confirm a saved shortlist instead. |
| The grids didn't add up (Step 32, below)                 | "Run stopped: the grids didn't add up to the book, so nothing was written." | Nothing you answered caused it. Send the file named on the screen.        |

### Step 32 · What a stopped Run looks like

This can't be made to happen from the screens. On the walk it was forced, to see what the window says.

PocketBook

✓

Extract

Consumer book Q3.csv · 8,000 loans

✓

Set up

10 columns read

✓

Choose tests

Where the book bleeds

!

Answer in workbook

Run stopped

5

Run

Run stopped

Nothing was written. Your answers so far are kept.

Run stopped: the grids didn't add up to the book, so nothing was written. grid fico x channel: rows adds up to 7999 across the cells, but the book says 8000.

That is a fault in PocketBook, not in your answers. The details are in /tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67-4db589c3cc31/scratchpad/walk27/home2/.pocketbook/last-error.txt; send that file over to get it fixed.

Open the workbook

Run

**Correct screen:** "Run stopped", not "answers needed". The message says the grids didn't add up to the book, that nothing was written, and where the details are.